

Binary images

- Black and white, 1 bit images
- Represented as boolean data, true or false
- If you are forced to use 8-bit images, use 255 for true, 0 for false
- Binary images are very useful in computer vision
- It is also possible to use binary images in image processing for selective filtering. For instance blurring background objects while keeping foreground object sharp
- There are many methods applicable to binary images, we will discuss them next week

Segmentation

- There are many methods to convert a grayscale image to binary image
- Methods that are designed to reduce grayscale level (ie. quantization methods) are not suitable in most cases. Our aim with binary images is not to reduce number of bits but rather get a truth map. These methods are called segmentation methods
- Simplest method is thresholding
- You simply cut from a value. Any value above or below this threshold will be replaced with true, rest will be false
- This value depends on the image and the application
- It is possible to use adaptive threshold algorithm to automatically determine threshold value. This method is not the only way to detect threshold value

Adaptive Threshold Algorithm

- This is an iterative algorithm. In each successive iteration, algorithm gets closer to the final value
- Once the threshold value is not changed, the iterative process is terminated
- Works on grayscale images
- Initial threshold is the average intensity of the image
- Loop until two successive iterations produce the same value:
 - Get the average intensity of pixels that have an intensity lower than the current threshold, Avg_{false}
 - Get the average intensity of pixels that have an intensity higher or equal to the current threshold, Avg_{true}
 - New threshold is $\frac{Avg_{false} + Avg_{true}}{2}$

Color image segmentation

- A simple method to segment color images is to use a circular color system
- Idea is to select a color on the circular color system and in the image choose pixels with values that are close to the given color
- To improve the results of the segmentation, closeness should be defined separately on each channel. Hue should have smallest tolerance. Additionally, lightness and saturation should have percentage based tolerance while hue should have absolute tolerance.
- The reason to use absolute in hue channel is due to not having a difference between high and low values. A difference of 60 (a full major color shift, red to yellow) from 0 is as effective as difference from 240. Same does not apply to other channels. In lightness, a value of 25 (10%) at 0 is the difference between black and dark gray while the same at 200, has almost no effect. This is due to human tendency of perceiving light in logarithmic scale.
- The reason behind having a smaller tolerance in hue is due to a 10% change in brightness will have minimum impact on human perception of the color while a 10% (36deg) change in hue is higher than the difference between red and orange.
- For instance, you could find red balls in a scene using hue $0 \text{ deg} \pm 10 \text{ deg}$, brightness $0.5 \pm 20\%$, saturation $0.8 \pm 20\%$.